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QT Interval and Cardiac Restitution Ratio Complexity in Standardbred Racehorses From Rest to Maximal Effort: Insights Into Arrhythmia Risk

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ABSTRACT

Background: Sudden cardiac death is common in racehorses. Factors associated with the QT interval that could predispose to fatal cardiac arrhythmias are unknown. Cardiac restitution, expressed as a ratio of QT/TQ, has been used in humans to assess arrhythmia risk but has not been described in horses during maximal intensity exercise.

Hypothesis/Objective: Describe factors associated with the QT interval and cardiac restitution ratio (CRR) in clinically normal Standardbred racehorses under race-day conditions.

Animals: Archival electrocardiograms from 42 Standardbred horses during live racing in Ontario.

Methods: Observational study performing an automated cardiac restitution analysis. Cardiac cycles were obtained from rest, non-race exercise, non-race recovery, live racing, and post-race recovery periods. Multivariable linear regression analysis was performed with both QT interval and CRR as outcomes of interest.

Results: Analysis of 3827 sequential pairs of cardiac cycles was performed. Exercise period and RR interval were highly associated with both QT interval and CRR. Other significant associations varied by exercise period and included: racing gait, sex, age, whether the horse received furosemide, and whether the horse experienced complex ventricular arrhythmias after racing. Interactions between gait and furosemide, and sex and gait were also significant.

Conclusions and Clinical Importance: An automated cardiac restitution analysis is feasible in exercising racehorses. The QT-RR interval relationship is multifactorial, and there are numerous significant associations that must be considered to interpret changes in QT interval and CRR in horses.

1 | Introduction

Exercise-associated sudden cardiac death (EASCD), most often attributed to fatal arrhythmias, is the second most common cause of death in racehorses [1–3]. These deaths fall inconsistently under several terms, including: sudden death [4], exercise-associated sudden death [5], EASCD [6], and sudden athletic death [7]. There is particular interest in the risk of EASCD in racehorses due to concerns regarding social license to operate. EASCD among equine athletes most often occurs either during

Abbreviations: CRR, cardiac restitution ratio; CVA, complex ventricular arrhythmias; EASCD, exercise-associated sudden cardiac death; iHR, instantaneous heart rate.

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or immediately after exercise and has been associated with both racing and training [6]. Fatal cardiac rhythm disturbances are rarely confirmed electrocardiographically [8], and therefore there is interest in performing routine cardiac rhythm monitoring to increase the likelihood of detection [9].

"QT interval" defines the period from the onset of the QRS (ventricular activation) to the end of the T-wave (ventricular repolarization) on the body surface ECG and approximates the overall duration of ventricular electrical activity. The QT interval is of interest in the study of arrhythmogenesis since ion fluxes during depolarization and repolarization need to be restored between successive action potentials to maintain electrical homeostasis and myocardial function. Although QT intervals are commonly "corrected"' for heart rate using one of various formulae [10], the QT-RR relationship is complex and multifactorial. A formulaic approach to heart rate correction is problematic, particularly during exercise where both QT and RR intervals are affected by autonomic balance [11]. Alternative methods of interpreting changes in the QT interval in exercising horses need to be explored.

Cardiac restitution refers to the relationship between QT and TQ (the interval between the end of the T-wave and start of the following QRS complex) and relates to the ability of cardiomyocytes to recover between successive action potentials [12]. A cardiac restitution ratio (CRR) has been described where the QT interval duration is divided by the preceding TQ interval duration, and this ratio is used as a measure of arrhythmia risk in humans; a ratio of greater than one is described as pro-arrhythmic [12, 13]. Cardiac restitution is reported in a small number of Thoroughbred racehorses during submaximal exercise [13]. However, cardiac restitution has not been described at race-intensity exercise or with temporal associations to live racing.

In addition to the need for an alternative expression of the QT-RR relationship, many previous studies examined electrocardiographic indices only at periods of stable heart rates [10, 13]. This practice does not consider what occurs during routine exercise with heart rates constantly fluctuating in response to demand and environmental stimuli. Measurements of electrocardiographic indices are frequently performed using either manual [14] or electronic calipers [10, 15, 16], which are subject to both intra- and interobserver variability [10, 16]. Therefore, the objectives of this study were to assess the feasibility of automatic electrocardiographic interval measurements and to describe factors associated with the QT interval and CRR in clinically normal Standardbred racehorses under race-day conditions. We hypothesize that automatic measurements will be possible using static (non-artificial intelligence) software algorithms. Additionally, we hypothesize CRR will exceed the previously described arrhythmogenic threshold at maximal intensity exercise and that CRR will be highest when heart rates are highest.

2 | Materials and Methods

2.1 | Data Collection

Forty-two archival digital ECGs collected from individual Standardbred racehorses during regularly scheduled racing were selected. The original ECGs were a nonrandom sample and involved all horses in each sampled race on two races per sampling day [17]. As such, the subset of horses published herein cannot be considered a random sample and contains an equal distribution of sexes (stallions, geldings, and females) and racing gait (pace, trot), hereafter referred to simply as gait. All ages are represented between 2 and 10 years, inclusive.

The continuous ECGs included time waiting in the paddock, warm-up(s), live racing, and the immediate race recovery period (up to 11 min after the end of the race). Collection details can be found elsewhere [17]. Electrocardiograms used were previously published describing rhythm variations during recovery from live racing [17]. Briefly, horses were instrumented with two bipolar modified base-apex leads as they arrived in the race paddock and were recorded until harness was removed after completion of the race. Age, sex, gait, and whether furosemide was administered were obtained for each horse from race data. Whether horses were previously categorized as having complex ventricular arrhythmias (CVAs) in the post-race period was obtained from archival data previously reported [17].

The original signal sampling frequency of 128 Hz was increased to 2000 Hz by simple linear interpolation [18] to increase the range of options by which algorithms could recognize fiducial points. Low amplitude high frequency noise and baseline drift were reduced by treating the data with a 20-point moving average smoothing function and by subtraction of a 500-point smoothing function, respectively. Custom software was used, as described previously [17], to achieve accurate QRS recognition. A custom algorithm recognizing QRS complexes by the maximal negative slope of the QRS was applied, and ECGs were manually inspected to confirm correct QRS recognition. Any recognition errors were corrected by reapplying the QRS recognition algorithm to the short ECG segments containing the recognition error. Individual interbeat intervals were manually selected in relatively noise-free segments for automated restitution analysis. Cardiac cycles that were clearly ectopic in origin due to differences in QRS morphology and those preceded by abnormally long TQ intervals (as observed in dropped beats and punctuated deceleration [17]) were not included. Each pair of cardiac cycles subjected to restitution analysis was visually inspected in OriginPro 2024b and cycles where the presence of noise/movement artifact caused fiducial points to be recognized at locations inconsistent with the literature [10, 13], were excluded. No manual corrections were performed to avoid introducing bias into the automated analysis.

For each pair of sequential cycles selected, software algorithms were used to identify points corresponding to the beginning and end of each waveform, therefore allowing measurement of the durations of the QRS complex, QT, TQ, and RR intervals. Instantaneous heart rates (iHR) were calculated from the RR interval and expressed as beats per minute (bpm), where applicable. When assessing the QRS complex and QT interval, only those for the first cycle in each selected pair of cycles were used to ensure no duplicated cycles. However, when concerning CRR, the QT interval of the second cardiac cycle in a pair was divided by the preceding TQ interval, consistent with the literature [13]. Relationships between variables

were visually examined and reported in figures generated in OriginPro 2024b.

2.2 | Categorization of Exercise Periods

Graphs of iHR over time were used to categorize ECG segments into one of five exercise periods according to the following descriptions.

Rest: Stationary in the paddock with a resting heart rate; includes transient increases in heart rate consistent with responses to environmental stimuli.

Non-Race Exercise: All sustained elevations in heart rate consistent with exercising that were not live racing include periods of acceleration, relatively stable heart rate, and gradual decreases in heart rate, consistent with slowing of exercise. Includes various levels of exercise, including one or more warmups, the post-parade, and horses getting to the gate at the beginning of the race.

Non-Race Recovery: Rapid deceleration of heart rate consistent with cessation of exercise when preceded by non-race exercise.

Live Racing: Due to the nature of the moving start in Ontario Standardbred racing, race start was defined by identifying the race end as the point where the heart rate began to rapidly decelerate consistent with cessation of exercise and subtracting the horse's own race time obtained from individual race data.

Post-Race Recovery: Rapid deceleration of heart rate consistent with cessation of exercise when preceded by live racing.

2.3 | Statistical Analysis

Statistical analyses were performed using SAS Version 9.4. Descriptive statistics (median, interquartile range, and maximum and minimum values) were generated for iHR, QRS duration, QT interval, and CRR for each exercise period. Multivariable linear regression was performed following a log transform of both QT interval and CRR as outcomes. Exercise period, sex, and gait were treated as categorical variables. Additionally, both furosemide status and CVA status were treated as binary variables with no discrimination between dosages or types of arrhythmias. Age and RR interval were treated as continuous variables. A random horse effect by sex, gait, furosemide status, and arrhythmia status was included. All main effects, two-way interactions, and second-order effects for continuous variables were included. Non-significant ($p > 0.05$) terms were sequentially removed beginning with the highest p value. Removed terms were re-introduced one at a time to ensure they remained non-significant. Higher order interactions were introduced as indicated based on the significance of multiple two-way interactions involving the same independent variable. In accordance with recent perspectives on reporting significance in veterinary medicine [19], results for terms remaining in the model were back-transformed and presented as ratios of medians with confidence intervals. Median differences (geometric means) were calculated by subtracting the median of one group of interest from that of another when there was a statistically significant difference between the groups.

3 | Results

The dataset contained cardiac cycles collected from 42 individual horses: 21 each of pacers and trotters; 14 each of geldings, stallions, and females; 13 horses that received furosemide; and 16 horses that were previously categorized as having CVA during the post-race recovery. The distribution of ages was 2 (3), 3 (13), 4 (5), 5 (4), 6 (7), 7 (5), 8 (2), 9 (1), and 10 years (2). The number of pairs of cardiac cycles obtained from each horse ranged from 60 to 185, with an average of 93 cycles per horse and a total of 3827 sequential pairs of cardiac cycles.

3.1 | Exercise Period

The distributions of QT interval, TQ interval, and CRR over the range of RR intervals are shown in Figure 1A–C and Figures S1 and S2. Additionally, QT interval expressed as a percentage of RR interval is shown in Figure 1D. The relationship between QT and RR is comprised of two components where the slope is more shallow at RR intervals longer than approximately 800 ms (iHR 75 bpm), and steeper at RR intervals shorter than 800 ms (Figure 1A).

Exercise period influenced the relationship between QT and RR interval (Figure 1A) with cardiac cycles occurring during non-race exercise having longer QT intervals for the same RR interval than cycles occurring during both non-race and post-race recovery. The shortest RR intervals (and therefore highest iHR) corresponded to the shortest QT intervals and occurred during live racing. The relationship between TQ and RR interval (Figure 1B, Figure S2) is also influenced by exercise period with cardiac cycles occurring at rest having shorter TQ intervals than those during non-race recovery at the same RR interval. These influences of exercise period on QT and TQ intervals are more pronounced when QT interval is expressed as a percent of RR interval (Figure 1D).

CRRs exceeded the arrhythmogenic threshold value of 1 the entire time horses were exercising (both non-race and live racing), and almost all of the recovery periods (both non-race and post-race, Figure 1C, Figure S1) [12]. At the same RR interval, there was variability in both the QT and TQ intervals, particularly evident when QT is expressed as a % of RR interval (Figure 1D). The contribution of QT interval to RR interval varied by approximately 20% at RR intervals less than 800 ms (iHR greater than 75 bpm); whereas there was roughly half the variability (~10%) at RR intervals greater than 800 ms.

The distribution of QT interval, TQ interval, iHR, and CRR relative to the race start time is shown in Figure 2 and Figures S3–S6. There was abrupt shortening of the QT interval at the beginning of the race (time = 0) and clear lengthening immediately at the end of the race (Figure 2A). TQ follows a

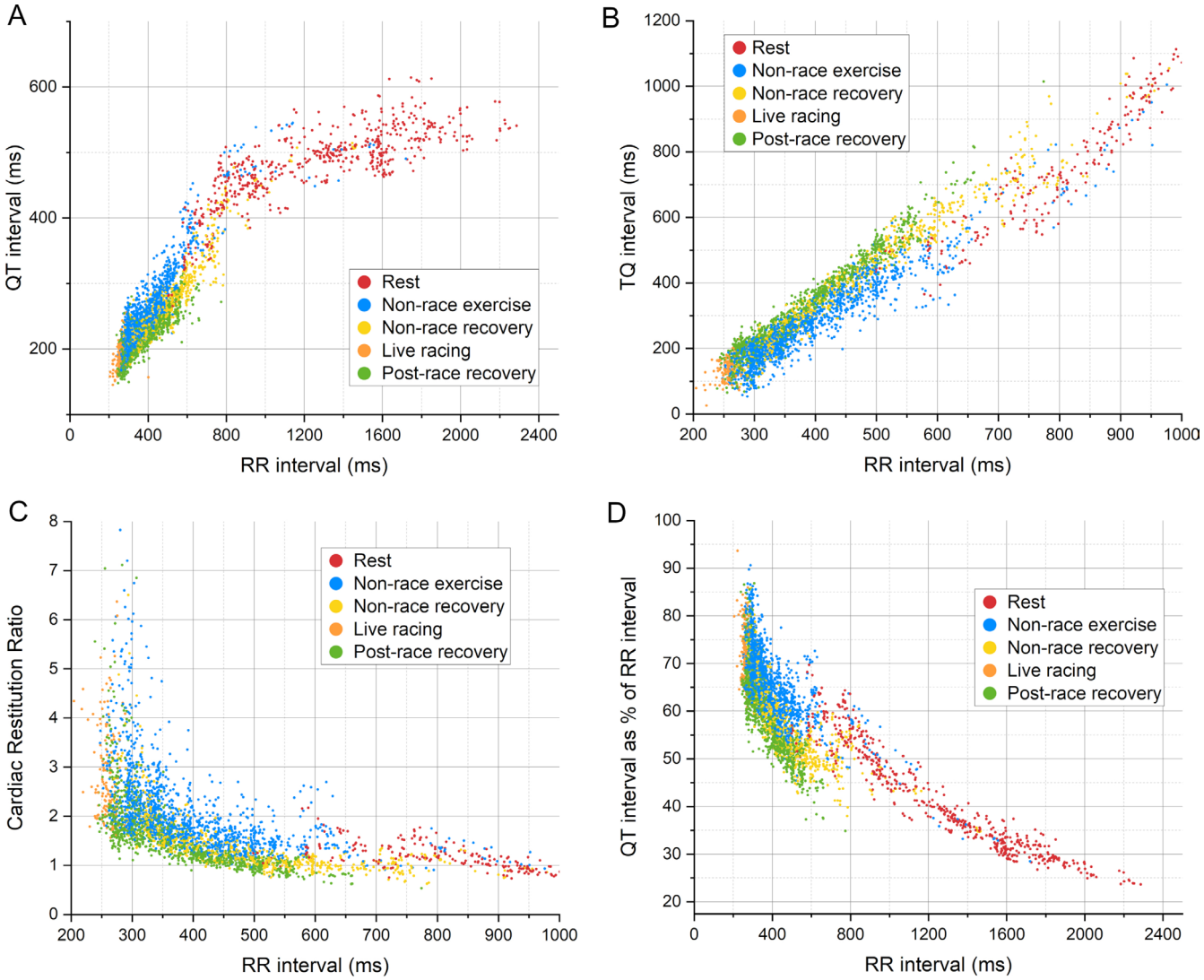

FIGURE 1 | QT interval (A), TQ interval (B), cardiac restitution ratio (C), and QT interval expressed as a % of RR interval (D) relative to RR interval by exercise period. The graphs shown in (B and C) are cropped images to highlight the data-dense areas which occur at higher heart rates, full graphs available as Figures S1 and S2.

similar pattern but lengthened at a steeper rate following the race (Figure 2B). Notably, during the post-race recovery period, the highest CRRs occur during the first 30 s after race completion (Figure 2D). There was marked variability in the QT duration and CRR leading up to the race due to differences in iHR (Figure 2C, Figure S5).

Instantaneous heart rates, as well as durations of QRS and QT interval within each exercise period are presented in Figure 3 with numerical data reported in Table S1. Similarly, CRR within each exercise period is presented in Figure 4 and Table S1. The median iHR was lowest at rest, as expected, and higher during live racing than during non-race exercise. Median iHRs are reported for non-race and post-race recovery; however, values have limited utility due to the fact that heart rate is changing (decreasing) in these periods. There were differences in median QRS duration between exercise periods, with rest having the longest, non-race exercise and non-race recovery being similar, and live racing and post-race recovery having slightly shorter durations. Median QT interval duration was longest at rest, followed by non-race recovery, non-race exercise, post-race recovery, and shortest during live racing. Median CRR (Figure 4) was highest during the race, followed by non-race exercise, post-race recovery, non-race recovery, and rest. Median CRR at rest was below the previously described arrhythmogenic threshold of 1; however, the range extends more than 2× above this limit. Lower limits of the interquartile ranges for CRR during non-race exercise, non-race recovery, live racing, and post-race recovery exceed 1; therefore, less than 25% of all cardiac cycles in these exercise periods were below the theoretical arrhythmogenic threshold. A statistical comparison for these data between exercise periods was not performed because a univariable analysis would oversimplify these data and ignore the effects of confounding variables identified below.

As noted in Figures 3 and 4, 24 horses had cardiac cycles of adequate quality for analysis during the live racing segment. Of the 18 horses in which there were no usable cardiac cycles, 13 had motion artifact throughout the race that precluded any attempt to measure electrocardiographic intervals. The remaining five horses had between 4 and 10 cardiac cycles which were

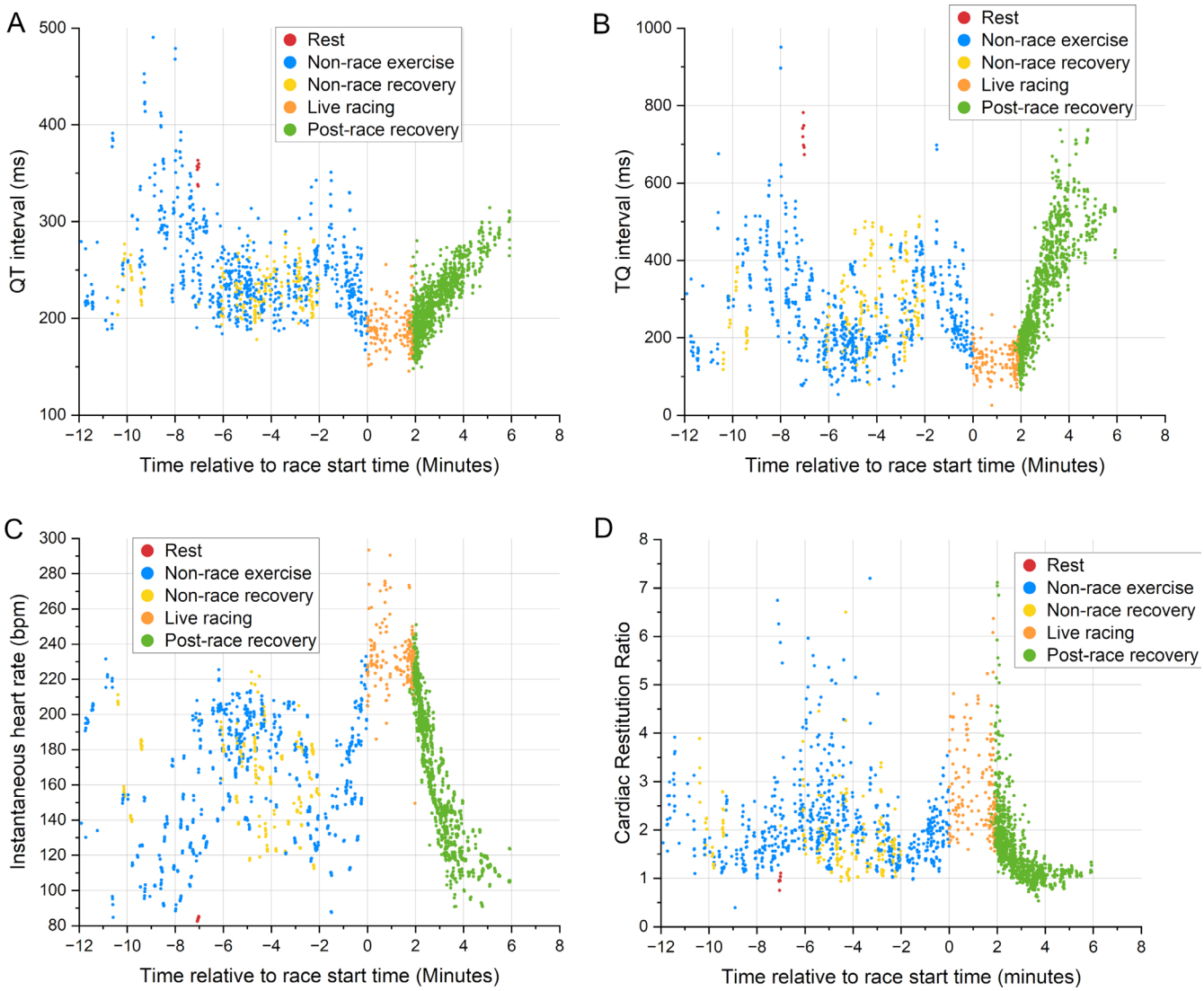

FIGURE 2 | QT interval (A), TQ interval (B), instantaneous heart rate (C), and cardiac restitution ratio (D) relative to race start time (time = 0) by exercise period. All graphs are cropped to highlight the peri-race areas, full graphs available as Figures S3–S6.

attempted, but the presence of noise/movement artifact interfered with reliable recognition of fiducial points, and therefore cycles were excluded from analysis.

3.2 | Multivariable Analyses

Multivariable analyses of variables associated with QT interval and with CRR revealed many independent variables and two-way interactions to be significant as well as several higher order interactions. This complexity resulted in difficulty interpreting biological meaning and raised concerns regarding model overfitting. Exercise period was highly significant under all circumstances and was involved in significant interactions with all other variables. Therefore, steps were taken to simplify the models. Ultimately, it was elected to model QT interval and CRR within each exercise period separately and to only include main effects, two-way interactions among categorical variables, and second-order effects for continuous variables. Ratios of medians for significant interactions were used with confidence intervals to compare magnitudes of effects and are presented in Tables 1 and 2. While these ratios indicate the magnitude and direction of the effect, it is difficult to ascertain biological relevance. Therefore, calculated median differences are also reported. However, due to the use of a logarithmic scale, confidence intervals could not be calculated for the reported medians and are only available for the ratios of medians (presented in Tables 1 and 2). QT interval and CRR were both significantly associated with RR interval as both linear and quadratic effects in all exercise periods. Individual equations can be calculated from the coefficients found in Tables S2–S11. Aside from RR interval, the factors associated with QT interval and CRR varied by exercise period. RR interval was the only factor significantly associated with either QT interval or CRR in the live racing period.

3.3 | Rest

During the rest period, gait was significantly associated with QT, with longer QT intervals occurring in pacers than trotters (median difference 33 ms). At rest, there was also a significant

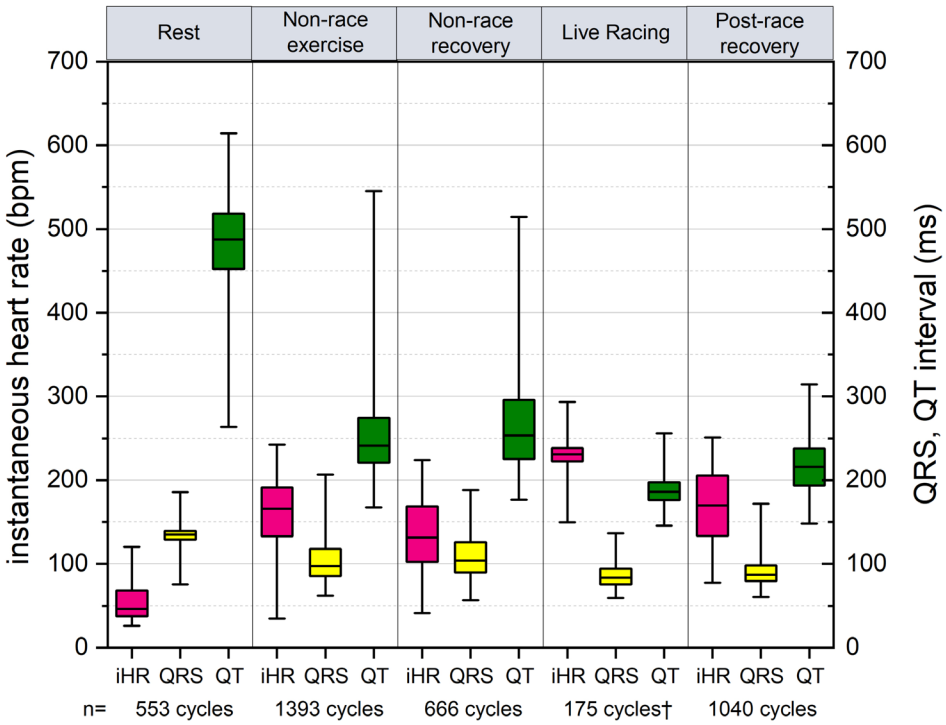

FIGURE 3 | Instantaneous heart rates (iHR) and electrocardiographic intervals by exercise period for cardiac cycles from 42 horses. Boxes show the interquartile range, whiskers show the minimum and maximum values, and horizontal lines represent the medians. Number of cycles (*n*) beneath each group. bpm, beats per minute; ms, milliseconds. †Eighteen horses did not have any pairs of cardiac cycles during the live racing segment of sufficient quality to perform automated analysis.

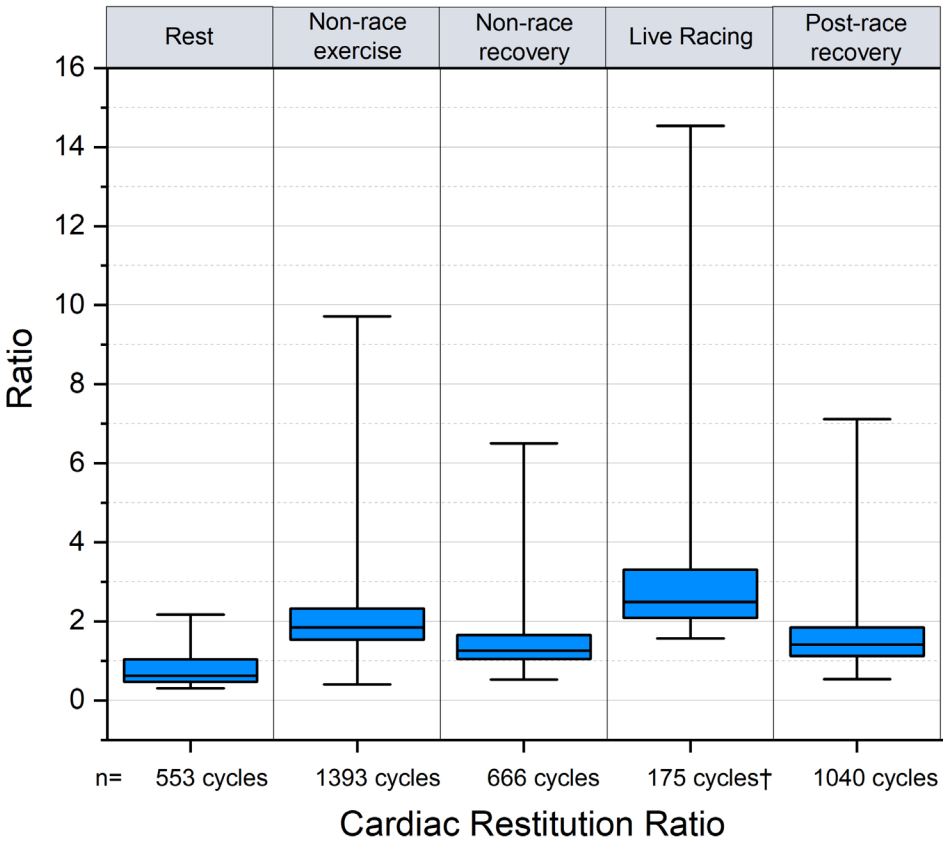

FIGURE 4 | Cardiac restitution ratio by exercise period for cardiac cycles from 42 horses. Boxes show the interquartile range, whiskers show the minimum and maximum values, and horizontal lines represent the medians. Number of cycles (*n*) beneath each group. †Eighteen horses did not have any pairs of cardiac cycles during the live racing segment of sufficient quality to perform automated analysis.

TABLE 1 | Factors significantly associated with QT interval while considering main effects, two-way interactions among categorical variables, and second-order effects for continuous variables.

Exercise period	Effect		Ratio of means	Confidence interval	*p*
Rest	Gait	P vs. T	1.07	1.04–1.11	< 0.001
	RRI (+)				< 0.001
	RRI × RRI (–)				< 0.001
	Gait × furosemide				
		P N vs. T Y	1.08	1.02–1.14	0.01
		P Y vs. T N	1.06	1.02–1.10	0.004
		P Y vs. T Y	1.12	1.06–1.18	< 0.001
		T N vs. T Y	1.05	1.00–1.11	0.05
Non-race exercise	Sex	F vs. S	0.97	0.95–1.00	0.04
	Gait	P vs. T	1.03	1.01–1.05	0.01
	Furosemide	N vs. Y	1.04	1.01–1.06	0.02
	Age (+)				0.02
	RRI (+)				< 0.001
	RRI × RRI (–)				< 0.001
	Sex × gait	F P vs. G P	0.94	0.91–0.98	0.001
		F P vs. S P	0.96	0.92–0.99	0.01
		F T vs. G P	0.96	0.93–1.00	0.03
		F T vs. G P	1.05	1.01–1.09	0.02
		G P vs. G T	1.09	1.05–1.14	< 0.001
		G T vs. S P	0.93	0.89–0.97	< 0.001
		G T vs. S T	0.95	0.91–0.99	0.01
Non-race recovery	Sex	F vs. G	0.95	0.92–0.98	0.002
		F vs. S	0.97	0.94–1.00	0.03
	RRI (+)				< 0.001
	RRI × RRI (–)				< 0.001
Live racing	RRI (+)				0.002
	RRI × RRI (–)				0.002
Post-race recovery	CVA	N vs. Y	1.03	1.00–1.06	0.04
	RRI (+)				< 0.001
	RRI × RRI (–)				< 0.001

Note: For categorical variables, the term first referenced is the numerator in calculation of ratios and only pairs that are statistically significant at $p < 0.05$ are shown. The direction of the relationship for continuous variables is represented by (+) for a positive association, and (–) for a negative association.
Abbreviations: CVA, complex ventricular arrhythmias; F, female; G, gelding; N, no; P, pacer; RRI, RR interval; S, stallion; T, trotter; Y, yes.

interaction between gait and furosemide administration. Among horses receiving furosemide, pacers had longer QT intervals than trotters (median difference 53 ms). Among trotters, those not receiving furosemide had longer QT intervals (median difference 24 ms) than those receiving furosemide. At rest, there was no significant difference between the QT intervals of pacers with or without furosemide administration.

At rest, gait was also significantly associated with higher CRRs in pacers than trotters (median difference 0.06). Age was

TABLE 2 | Factors significantly associated with cardiac restitution ratio while considering main effects, two-way interactions among categorical variables, and second-order effects for continuous variables.

Exercise period	Effect		Ratio of means	Confidence interval	*p*
Rest	Gait	P vs. T	1.08	1.03–1.14	0.002
	Age (+)				0.03
	RRI (−)				< 0.001
	RRI × RRI (+)				< 0.001
Non-race exercise	Sex	G vs. S	0.88	0.79–0.98	0.02
	Furosemide	N vs. Y	1.18	1.05–1.33	0.01
	Age (+)				0.03
	RRI (−)				< 0.001
	RRI × RRI (+)				< 0.001
	Sex × gait	F P vs. G P	0.87	0.75–1.00	0.05
		F P vs. S P	0.86	0.75–0.99	0.04
		F T vs. G T	1.24	1.06–1.45	0.01
		G P vs. G T	1.32	1.12–1.56	0.002
		G T vs. S P	0.75	0.64–0.88	< 0.001
		G T vs. S T	0.78	0.66–0.92	0.004
Non-race recovery	Sex	F vs. G	0.87	0.80–0.95	0.002
		F vs. S	0.89	0.82–0.97	0.01
	RRI (−)				< 0.001
	RRI × RRI (+)				< 0.001
Live racing	RRI (−)				0.003
	RRI × RRI (+)				0.02
Post-race recovery	CVA	N vs. Y	1.11	1.02–1.20	0.02
	Age (+)				0.05
	RRI (−)				< 0.001
	RRI × RRI (+)				< 0.001

Note: For categorical variables, the term first referenced is the numerator in calculation of ratios and only pairs that are statistically significant at $p < 0.05$ are shown. The direction of the relationship for continuous variables is represented by (+) for a positive association, and (−) for a negative association.
Abbreviations: CVA, complex ventricular arrhythmias; F, female; G, gelding; N, no; P, pacer; RRI, RR interval; S, stallion; T, trotter; Y, yes.

significantly associated with CRR, with CRR increasing as age increased. Intercepts can be found in Table S7.

3.4 | Non-Race Exercise

As for the rest period, gait was significantly associated with QT interval during exercise, with longer QT intervals in pacers than trotters (median difference 7 ms during non-race exercise). Females had shorter QT intervals than stallions (median difference 6 ms). Geldings were not statistically different from either females or stallions. Horses not receiving furosemide had longer QT intervals (median difference 9 ms) than those receiving furosemide. During non-race exercise, there was a significant interaction between sex and gait. Among pacers, females had the shortest QT intervals (median difference 11 ms from stallions and 15 ms from geldings). Among trotters, geldings had the shortest QT intervals (median difference 13 ms from stallions and 12 ms from females). This contrast appears to be because geldings had shorter QT intervals when they were trotters versus pacers (median difference 22 ms).

Despite the differences in QT interval for gait, there was no significant difference in the CRR between pacers and trotters. Geldings had lower CRRs than stallions (median difference 0.25) with no significant difference in CRR between females and either stallions or geldings. Horses not receiving

furosemide had higher CRRs (median difference 0.33) than those receiving furosemide. There was a significant interaction between sex and gait. Among pacers, females had the lowest CRRs (median difference 0.29 for stallions and 0.28 for geldings). Among trotters, geldings had the lowest CRRs (median difference 0.45 for stallions and 0.38 for females). There was no significant difference between stallion and gelding pacers, and female and stallion trotters. As for the finding for QT interval, this contrast appears to be because geldings had lower CRRs when they are trotters versus pacers (median difference 0.52). Age was significantly associated with CRR, with higher CRRs occurring as age increased. Intercepts can be found in Table S8.

3.5 | Recovery Periods

Aside from RR interval, the only significant association during the non-race recovery period was sex, with females having the shortest QT interval (median difference 9 ms from stallions and 14 ms from geldings). There was no significant difference between stallions and geldings. Interestingly, this same association with sex was not found during the post-race recovery period, with CVA classification as the only significant categorical variable. Horses who experienced CVA in the immediate post-race period had shorter QT intervals (median difference 6 ms) than those who did not.

Sex was significantly associated with CRR during the non-race recovery period only for females, who had the lowest CRR (median difference 0.18 for geldings and 0.15 for stallions). There was no significant difference between geldings and stallions. During the race recovery and similar to the QT interval, horses that experienced CVA in the immediate post-race period had lower CRRs (median difference 0.15) than those who did not. Age was also significantly associated with CRR in the post-race recovery, with higher CRRs occurring as age increased. Intercepts can be found in Table S11.

4 | Discussion

Consistent with the hypothesis, CRR exceeded the theoretical arrhythmogenic threshold of 1 during live racing, as well as much of the time during non-race exercise and recovery from exercise. Additionally, an automated approach to electrocardiographic measurements was feasible using archival ECGs from Standardbred racehorses. The hypothesis that CRR would be highest when heart rates were highest (live racing) was also confirmed. However, there were sporadically high CRRs during periods of non-racing exercise, which may represent periods of elevated risk. While SCD associated with live racing receives the most attention, there is evidence that it is more frequently associated with training exercise [6]. However, there are no studies that account for the frequency of training exercise versus live racing; therefore, the risk of SCD during different types of exercise remains unknown. The observed sporadic elevations in CRRs during non-race exercise are an interesting finding, and future studies including controlled periods of acceleration and deceleration are needed to specifically assess this phenomenon. One possible explanation is that there is a delay in response of the QT interval. In humans at rest, a 2-min delay in response of the QT interval to changes in RR interval has been described [20]. However, this phenomenon, termed *QT hysteresis* or *cardiac memory*, was not evident in the present data. An immediate lengthening of QT and TQ intervals at the end of the live racing period was found, suggesting that QT hysteresis does not occur to any significant extent in racehorses during maximal intensity exercise or recovery.

The present study highlights the complexity of the QT-RR interval relationship. Additionally, variability in QT and TQ intervals was identified at the same RR intervals, with greater variability at RR intervals less than 800 ms (iHR greater than 75 bpm). The QT interval on a body surface ECG is a function of both the action potential duration and the synchronicity of ventricular activation and repolarization. Therefore, variability in the QT interval may reflect differences in the activity of ion channels resulting in changes in the action potential and/or conduction velocity at the cellular level. An alternative explanation includes variation in conduction pathways and possible conduction blocks, which may change depending on the RR interval, sympathetic stimulation, or catecholamine response. Any of the aforementioned explanations could predispose to re-entry arrhythmias in performance horses and are worthy of further investigation.

The relationship between QT interval and RR interval in the present study is consistent with the segmental relationship reported previously in Standardbred horses [10] and differs from that reported in humans [21]. One possible explanation for this difference is that humans have a narrower range of observed heart rates and therefore do not have the opportunity to demonstrate the shallower slope of the QT-RR relationship seen at lower heart rates. However, an alternative hypothesis is that this segmental relationship is unique to horses and could represent adaptation of horses to aerobic exercise. Comparative investigations of the QT-RR relationship at a wide range of heart rate ranges may yield additional insight. The bending point of the QT-RR relationship in horses was previously reported as approximately 850 ms (iHR of 70 bpm) [10] and appears similar in the present report at approximately 800 ms (iHR of 75 bpm). However, changes in the QT-RR interval relationship with exercise period are a novel finding in horses and are consistent with work in humans demonstrating shorter QT intervals during recovery that were ascribed to the return of parasympathetic activity [21]. Additionally, the incongruency of the TQ-RR interval relationship between cardiac cycles at rest and those during exercise and recovery is a novel observation and suggests some influence on the TQ interval that is not currently accounted for at RR intervals in the 600–800 ms range. Additionally, it should be considered that the factors which influence the length of QT and TQ may differ, offering a possible explanation for why some of the factors associated with CRR were not associated with QT (age at rest and during post-race recovery).

Linear regression analysis revealed horses experiencing CVA during the immediate post-race period had shorter QT intervals and lower CRRs than those unaffected. This finding was unexpected and cannot be explained within the present report. One possible explanation is that these horses have higher parasympathetic stimulation than their counterparts and therefore

shortening of the QT interval in the recovery period described above is more pronounced in these horses. It is possible, therefore, that CRRs may be predictive of arrhythmia risk, but not in the manner originally suspected. A focused beat-by-beat analysis investigating changes in CRR during periods of punctuated deceleration and CVA is currently underway and should yield additional insight.

Our findings support previous reports that there are sex differences in the length of QT interval in horses [10], however suggests a sex by gait interaction must be considered. In both the present and previous studies, stallions had longer QT intervals than females at the same RR interval [10]. However, unlike previous reports [10], we found no difference between geldings and stallions. In the previous study, horses classified as "Standardbred American light type" were not distinguished as either pacers or trotters. However, we found the median QT interval length for stallions was higher than gelding trotters and lower than gelding pacers, providing an explanation for the lack of statistical difference between stallions and geldings. The reason for a sex–gait interaction in Standardbred horses could be related to genetic differences between the breed subsets [22] but requires further investigation. Additionally, it is interesting that an interaction between sex and gait was significant only in the non-race exercise period. This finding may highlight important differences in type of non-race exercise performed (duration, intensity) depending on the horse's signalment and warrants further investigation in the context of arrhythmia risk.

Our finding of shorter QT intervals in horses receiving furosemide is consistent with previous reports [14]. However, it should be noted that the race data used in the present study only documents horses receiving furosemide on race day and cannot exclude the possibility that horses in the "no furosemide" group may have been previously medicated. Despite previous results being consistent with our findings, caution should be used when comparing measurements before and after exercise due to the effect of the exercise period on the QT interval, as was done in the referenced study [14]. Our results suggest that QT comparisons need to be much more standardized and highly controlled to minimize the impact of confounding variables due to the complexity of the QT-RR relationship. When looking at the factors associated with CRR, our results are consistent with previous studies in Standardbred racehorses reporting no increased risk of arrhythmias in horses receiving furosemide [17, 23]. However, the present observational study is unable to determine the actual effect of drugs administered and can only make comparisons between groups of horses. Randomized controlled studies would be needed to determine whether there is a decreased risk of cardiac arrhythmias with furosemide administration due to shorter QT intervals and lower CRRs.

The multivariable analysis demonstrated the complexity of the associations between QT and the tested independent variables with many significant terms. Some associations were consistent across exercise periods (RR interval) while others differed between exercise periods. While the median (equivalent to the geometric mean) differences are reported here to give some idea of the magnitude of the differences, these values are an approximation due to the simplified statistical analysis, which was necessary with the complexity of the data. There are surely other significant factors, simple and higher order interactions, which could not be investigated in the present report. Due to the motion artifact that occurs with exercising ECGs and the short timeframe corresponding to live racing (roughly 2 min), there were fewer cardiac cycles available for analysis, and not all horses had ECG signals of adequate quality during the race for analysis. Therefore, while RR interval was the only statistically significant association during live racing, the power with which to detect associations was lower in this period. The possibility remains that there are other biologically relevant associations with QT interval and CRR during live racing that could not be determined in the present study. This limited availability of data during live racing is a limitation of the present study and an unfortunate universal challenge with ECGs during maximal effort, emphasizing the need for high-quality ECG recordings. Measuring electrocardiographic intervals requires even higher signal quality than rhythm analysis to confidently identify waveform boundaries. Despite using archival data not intended for interval measurement, the desired information was obtained during maximal effort from nearly 60% of recordings. Purpose-collected data using newer technology will likely yield even better results.

In conclusion, this report demonstrates the feasibility of an automatic electrocardiographic analysis which may facilitate large-scale investigation of electrocardiographic intervals in racehorses. We have demonstrated the complexity of the QT-RR interval relationship. Therefore, formulaic QT correction seems inappropriate for use in exercising horses. Some of the nuances of the factors associated with QT interval in Standardbred racehorses are described, including: exercise period, gait, sex, age, furosemide administration, CVA status, and interactions between gait and furosemide, and sex and gait. CRR is presented as a method to consider changes in both the QT and TQ intervals, and factors associated with CRR in Standardbred racehorses have been similarly described. CRR was found to be highest during live racing, the first 30 s after completion of the race, and sporadically during non-racing exercise, corresponding to periods when SCD frequently occurs. However, SCD was not an outcome in the current study; therefore, evidence that CRR can be used for a risk assessment in racehorses was not directly investigated. Horses with post-race CVA had shorter QT intervals and lower CRRs only during the post-race recovery period. The complexity of QT, TQ, and RR interval relationships suggests much more information is required to interpret changes in QT interval and CRR in horses.

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Disclosure

Authors declare no off-label use of antimicrobials.

Ethics Statement

Authors declare no institutional animal care and use committee, institutional animal untilzation protocol (AUP), or other approval was needed for archival data, however AUP #07R061 was approved by the University of Guelph for the original ECG collection. Authors declare human ethics approval was not needed.

Conflicts of Interest

The authors declare no conflicts of interest. The software used in the present study is intended to be published in the Biomedical Engineering literature and will be made available at no cost to researchers. There has not been, nor will there be financial gain from the software.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Figure S1:** Cardiac restitution ratio relative to RR interval in different exercise periods. This is the same graph as Figure 1B but here the full dataset is shown. **Figure S2:** TQ interval relative to RR interval in different exercise periods. This is the same graph as Figure 1C but here the full dataset is shown. **Figure S3:** QT interval relative to race start time (time = 0) in different exercise periods. This is the same graph as Figure 2A but here the full dataset is shown. **Figure S4:** TQ interval relative to race start time (time = 0) in different exercise periods. This is the same graph as Figure 2B but here the full dataset is shown. **Figure S5:** Instantaneous heart rate relative to race start time (time = 0) in different exercise periods. This is the same graph as Figure 2C but here the full dataset is shown. **Figure S6:** Cardiac restitution ratio relative to race start time (time = 0) in different exercise periods. This is the same graph as Figure 2D but here the full dataset is shown. **Table S1:** Instantaneous heart rates (iHR), electrocardiographic intervals, and cardiac restitution ratio (CRR) by exercise period. **Table S2:** Solutions for fixed effects of QT interval during the rest period. **Table S3:** Solutions for fixed effects of QT interval during the non-race exercise period. **Table S4:** Solutions for fixed effects of QT interval during the non-race recovery period. **Table S5:** Solutions for fixed effects of QT interval during the live racing period. **Table S6:** Solutions for fixed effects of QT interval during the post-race recovery period. **Table S7:** Solutions for fixed effects of cardiac restitution ratio during the rest period. **Table S8:** Solutions for fixed effects of cardiac restitution ratio during the non-race exercise period. **Table S9:** Solutions for fixed effects of cardiac restitution ratio during the non-race recovery period. **Table S10:** Solutions for fixed effects of cardiac restitution ratio during the live racing period. **Table S11:** Solutions for fixed effects of cardiac restitution ratio during the post-race recovery period.